

Shitao Jin

AI in Architecture | Collective Intelligence | Megaproject Decision-Making

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22 published papers • 16 first- or corresponding-author peer-reviewed journal articles • 298 citations • h-index / i10-index: 8 / 7

RESEARCH PROFILE

PhD candidate in Architecture with extensive experience applying computational decision-analysis and artificial intelligence (AI) methods, including multi-agent reinforcement learning, natural language processing, and fuzzy-grey decision models, to architectural programming, megaproject governance, and human-centered built-environment evaluation. Proven track record of developing evidence-based decision-support frameworks for early-stage problem definition, stakeholder coordination, risk assessment, public concern analysis, and value-based design evaluation in complex built-environment systems.

EDUCATION & VISITING RESEARCH

Tongji University, Shanghai, CHINA

2023 - 2027 (expected)

PhD in Architecture, Department of Architecture.

- Thesis: Collective-Intelligence-Based Collaborative Decision-making Systems for Architectural Programming
- GPA: 3.60/4.00 (Ranked in the top 5%)

The University of Melbourne, Melbourne, AUSTRALIA

2025 - 2026

Visiting PhD Researcher, Department of Infrastructure Engineering.

- Research focus: Value Management, Stakeholder Coordination, and Complexity Governance in Megaprojects

Huazhong University of Science and Technology, Wuhan, CHINA

2020 - 2023

M.Arch., Department of Architecture.

- Thesis: Design and Evaluation of Active-Learning Built Environments in Higher Education
- GPA: 3.50/4.00 (Ranked in the top 5%)

SELECTED PUBLICATIONS (* denotes corresponding author)

Shitao Jin*, "Dynamics and Influences Analysis of Public Concerns in Mega Construction Projects: An Integrated Topic and Sentiment Modeling Approach", *Journal of Construction Engineering and Management*, 2026, 152(8), 04026104.

Shitao Jin* & Xudong Miao, "Risk Identification and Assessment of Mega Construction Projects in Uncertain Environments: A Gray Fuzzy Group Decision-Making and Monte Carlo Simulation Method", *Journal of Architectural Engineering*, 2026, 32(2), 04026010.

Huijun Tu, **Shitao Jin***, *et al.*, "A value-based programming framework for enhancing children's well-being in urban communities", *Frontiers of Architectural Research*, 2026, 1-20.

Shitao Jin*, "Current status and trends of megaproject research: bibliometric and text mining analysis", *Engineering, Construction and Architectural Management*, 2026, 33(6), 4546-4573. [8 cites]

Shitao Jin*, "Measuring complexity in mega construction projects: fuzzy comprehensive evaluation and grey relational analysis", *Engineering, Construction and Architectural Management*, 2026, 33(1), 284-317. [7 cites]

Shitao Jin*, "Interdisciplinary perspective on architectural programming: current status and future directions", *Engineering, Construction and Architectural Management*, 2025, 32(11), 7332-7372. [15 cites]

Shitao Jin* & Yuan Deng, "Identification and evaluation of key stakeholders in mega construction projects: social network analysis and multi-criteria decision-making", *Engineering, Construction and Architectural Management*, 2025, 1-38. [1 cite]

Shitao Jin & Huijun Tu*, "Current status and research progress in architectural programming: A comparative analysis between China and other countries", *Frontiers of Architectural Research*, 2025, 14(5), 487-509. [6 cites]

Shitao Jin*, "The application of collective intelligence in the construction industry: a review of the current state, challenges, and opportunities", *Architectural Engineering and Design Management*, 2025, 21(5), 1091-1116. [6 cites]

Huijun Tu & **Shitao Jin***, "A group decision-making model for architectural programming in megaprojects", *Engineering, Construction and Architectural Management*, 2024, 31(13), 342-368. [11 cites]

Shitao Jin, Huijun Tu*, *et al.*, "Enhancing Architectural Education through Artificial Intelligence: A Case Study of an AI-Assisted Architectural Programming and Design Course", *Buildings*, 2024, 14(6), 1613. [77 cites]

CONFERENCE PAPERS

Shitao Jin*, Yuan Deng, & Huijun Tu, "Collective Intelligence in Architectural Programming: Multi-Agent Reinforcement Learning for Optimization in Mega Transportation Projects", *Proceedings of the 31st Computer-Aided Architectural Design Research in Asia (CAADRIA)*, 2026, Taiwan, China.

Shitao Jin*, Yuan Deng, & Huijun Tu, "Enabling Intelligent Collaboration in Architectural Programming: A Hierarchical Multi-Agent Reinforcement Learning Framework for Mega Transportation Projects", *Proceedings of Construction Research Congress (CRC)*, 2026, San Antonio, USA.

Shitao Jin*, Yuan Deng, & Huijun Tu, "Reprogramming Urban Communities: Enhancing Children's Well-being through a Value-Based Programming Approach in Inclusive Planning and Design", *Proceedings of the 60th ISOCARP World Planning Congress*, 2024, Siena, Italy.

TEACHING & MENTORING

The University of Melbourne, Melbourne, AUSTRALIA

Sep. 2025 - Jan. 2026

Teaching Assistant, Engineering Management Capstone (ENGM90016).

- Supported team-based construction cases, risk-aware project plans, budgets, and presentations.

Tongji University, Shanghai, CHINA

Sep. 2024 - Jan. 2025

Teaching Assistant, AI-Enabled Architectural Programming and Design.

- Guided child-friendly public-space briefs, LLM/AIGC workflows, and evidence-based studio deliverables; developed an AI-assisted course model.

ACADEMIC SERVICE

Verified peer reviewer for 169 manuscripts across 10 journals, including JCEM, ECAM, JAABE, JSSR, CAEE, *PLOS ONE*, and *Buildings*, among others.

RESEARCH & TECHNICAL SKILLS

- Computational methods:** Multi-agent reinforcement learning; machine learning; deep learning; LLM fine-tuning; topic and sentiment modeling.
- Decision analysis:** Group decision-making; multi-criteria decision-making; social network analysis; fuzzy/grey methods; Monte Carlo simulation.
- Design and research tools:** Rhino/Grasshopper; Revit; SketchUp; AutoCAD; ComfyUI; SPSS; NVivo.
- Languages:** English (fluent; IELTS 7.0); Mandarin Chinese (native).

AWARDS & HONORS

National Scholarship for Ph.D. Students, 2024 and 2025; Outstanding Student, 2024 and 2025;

National Scholarship for Master Students, 2022; Outstanding Graduate, 2023; Outstanding Student, 2022; First-Class Scholarship, 2020, 2021, and 2023.